Improving Molecular Property Prediction with Score-Based Models

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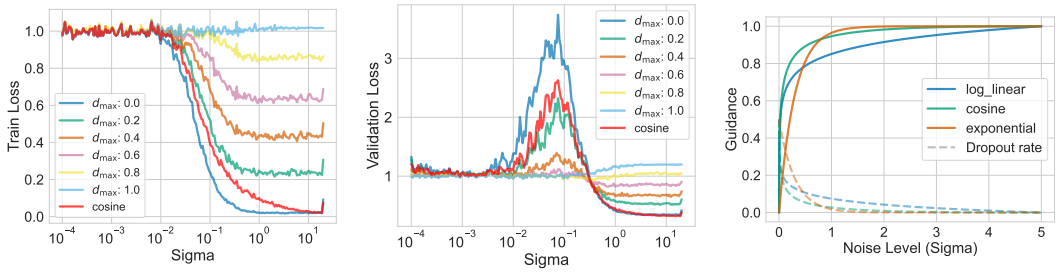
Abstract

We explore diffusion as a flexible framework for learning predictive models in molecular property prediction, highlighting the specific challenges and proposing tailored architectural and diffusion design elements for this domain. By viewing prediction as the time-reversal of a diffusion process, conditioned on input features, we achieve models that rival or surpass those trained with traditional maximum likelihood objectives. This approach inherently offers calibrated predictive uncertainties, task-conditioned predictions in multitask contexts, and resilience to class imbalances. Empirically, diffusion-based training improves performance across typical feature extractors and graph neural networks in small-molecule regression tasks.

1 Introduction

Improving the reliability of computational prediction has the potential to fundamentally transform the way molecules are designed, reducing experimental costs and speeding up the drug development process [6, 51]. Score-based models have gained prominence in generative modeling due to their inherent ability to model complex, multimodal distributions, while retaining sample fidelity on par with GANs [11]. This work explores these properties within a predictive context, adapting diffusion models to learn conditional distributions of properties given molecular featurizations. By applying diffusion to the space of molecular properties, rather than molecular structures, our approach enables novel conditioning mechanisms for multitask learning, where the prediction of one property can be informed by noisy observations from related tasks—a setting with potential to alleviate issues of data scarcity typical in molecular applications.

We thus propose DiffPred, a diffusion-based framework which we demonstrate is an effective paradigm for molecular property prediction. By exploring different design aspects of diffusion models in this context, we propose a joint training and sampling guidance schedule to improve the effectiveness of our approach at different noise levels. We additionally introduce an optional gating mechanism in the residual component of our architecture, enabling the model to selectively switch between feature-based or target-based information flow. Further, we evaluate the properties of score-based predictors, finding that they are capable of producing calibrated uncertainty estimates, agreeing with the intuition from generative modeling that diffusion models excel at covering the full support of a distribution while maintaining high fidelity. We further discuss applications for activity cliff prediction, a key challenge in molecular datasets, and multitask learning. Through empirical evaluation, DiffPred demonstrates improvements in both single task and multitask regression and classification benchmarks relative to leading graph neural network architectures, both on ADMET properties and activity prediction datasets.



(a) (b) (c)

Figure 1: Panels (a) and (b): Training and validation loss profiles across noise levels after training a DiffPred MLP for 500 epochs on CHEMBL228_Ki. We plot the effect of varying levels of guidance dropout with a constant guidance strength of 0.9, against a cosine schedule guidance with $d_{\max} = 0.5$. The cosine schedule reduces overfitting relative to no dropout at intermediate noise, while achieving similarly low loss on high noise levels to full guidance/zero dropout. Panel (c): Visual of guidance schedules and associated dropout rates for $d_{\max} = 0.5$ (more detailed ablations on schedule can be found in Appendix H).

2 Molecular Property Prediction with Diffusion Models

2.1 Feature-Conditioned Predictive Diffusion

We briefly recall the background on diffusion models in appendix B. As we wish to train a model to generate (possibly vector-valued) predictions, our objective will be to train a model to learn the *conditional* scores $\nabla_{y_t|x} \log p_t(y_t|x)$. Our implementation of this score-based paradigm follows the core guidelines detailed in the EDM [30] diffusion framework. The resulting SDE sets the noising coefficient $\sigma(t) = t$, such that the perturbation kernel has the simple form $p_{0t}(y_t|y_0) = \mathcal{N}(y_t|y_0, t^2\mathbf{I})$, thus removing any distinction between noise and time. Accordingly, we use both σ and t interchangeably, according to context, throughout this work.

Training is stabilized via σ-dependent input and output scaling factors. Furthermore, a noise-parameterized weighting of noisy target values y_t and network outputs $f_\theta(\cdot)$ smoothly interpolates between predicting additive noise and the original target values. This modulation of the prediction target is similar to v-prediction [47]. More formally, the resulting diffusion model output is given by

$$F_\theta(x, y_t, t) = \frac{\sigma_d^2}{t^2 + \sigma_d^2} y_t + \frac{\sigma_d t}{\sqrt{t^2 + \sigma_d^2}} f_\theta(x, \tilde{y}_t, \tilde{t}), \tag{1}$$

where $\tilde{y}_t = {}^{y_t}/_{\sqrt{t^2+\sigma_d^2}}$, $\tilde{t} = {}^{\ln(t)}/_4$, and σ_d corresponds to the uncorrupted data standard deviation. The function f_θ is a neural network, and its output is conditioned on input features x by adding them to the model's inputs. At high values of t, the first term involving y_t will resemble pure noise and its weighting will tend to zero. The network f_θ will thus tend to directly predict the clean target values. At low noise, y_t may be very close to y and this term will dominate the output. The network f_θ will then act as a typical score network in trying to estimate the additive noise ϵ_t which generated y_t from the clean data. The final denoising objective takes the form of a weighted L_2 loss

$$\mathbb{E}_{x,t,y_t}\left[\frac{t^2 + \sigma_d^2}{t^2 \sigma_d^2} \|F_\theta(x, y_t, t) - y\|_2^2\right] \tag{2}$$

with the expectation estimated via Monte-Carlo sampling. The loss weighting equalizes the expected magnitudes across noise levels. We sample from our model using the first-order stochastic sampler introduced in [30] (refer to Algorithm 2).

In unconditional diffusion, learning typically is only effective at intermediate noise levels. At high noise levels, where the input is indistinguishable from Gaussian noise, it is nearly impossible to recover the underlying signal, whereas at extremely low levels of noise, the network may struggle to disentangle the added noise from the signal.

This typical loss profile can be seen in Appendix H, Figure 7 (a) (dotted red curve), and has led to the proposal of sampling noise as a log-normal distribution [30] in order to maximize learning efficiency. However, an important consideration in the context of prediction tasks is how the presence of feature

information affects these training dynamics. Indeed, at high noise the features will now be highly informative in correctly predicting the raw target values, and the training dynamics should resemble those of a typical MSE-trained neural network. From this perspective, the high noise training regime may also be interpreted as training a noise-robust predictor [40]. Due to these characteristics, we do not discriminate between noise levels during training and sample σ_t from a log-uniform distribution $\ln(\sigma) \sim U(\ln(\sigma_{min}, \sigma_{max}))$, as in the variance exploding (VE) diffusion framework [53, 54].

2.2 Scheduled Guidance

In addition to guiding the model's predictions by directly conditioning on the features via the network inputs, we employ guidance techniques in order to modulate the degree to which the model is biased to an unconditional prior over the target distribution $p_0(y)$. Classifier-free guidance [23] is an inference-time technique which guides samples of the model towards conditioning information via a pre-defined weighting of the conditional and unconditional scores. Taking y as diffusion targets and x as the conditioning features, the classifier-free guidance expression is

$$s_\theta(y_t|x) = \gamma \nabla_{y_t} \log p_\theta(y_t|x) + (1-\gamma)\nabla_{y_t} \log p_\theta(y_t), \tag{3}$$

which flexibly mixes conditional and unconditional score information. In the predictive diffusion setting, we heavily favour learning the conditional score, as our primary objective is accuracy rather than diversity. However, because the conditional denoising at lower noise regimes may be particularly challenging, particularly when the ground truth function is itself noisy, we propose to smoothly decrease the guidance level as the sampling procedure progresses to lower noise levels. Specifically, we parameterize the guidance weight via a cosine schedule in log-space given by $\gamma(\sigma, \sigma_{min}, \sigma_{max}) = \frac{1}{2}\left(1 - \cos\left(\pi * \left(\frac{k}{K}\right)\right)\right)$, where $k = \log(\sigma) - \log(\sigma_{min})$ and $K = \log(\sigma_{max}) - \log(\sigma_{min})$ This rapidly approaches zero as $\sigma \to 0$, but maintains a strong guidance weight over most of the noise range during sampling. We experimented with alternative parameterizations, depicted in Figure 1(c), and found the cosine schedule to consistently perform the best among them (see Appendix H).

While this approach to dynamic guidance has been explored in recent work on conditional image generation [59, 7, 3, 16], our reasoning is different: instead of relaxing guidance early in sampling to improve sample diversity, our guidance weights are monotonically *decreasing* in order to reduce conditioning strength throughout sampling. In this regime, the dynamic guidance schedule effectively acts as an adaptive regularizer at low noise, which pushes samples towards the marginal distribution of the prior as training becomes more challenging near $\sigma = 0$.

During training, the unconditional score is typically trained by masking out the conditional information at a fixed dropout rate. We extend the idea of dynamic guidance to our training regime by further performing the masking of features at a timestep-dependent rate of $d_{\max} * (1 - \gamma(\sigma))$, thus increasing the training time provided to the unconditional score according to its expected contribution during sampling. We illustrate the role that an adaptive dropout rate may have during training and sampling in Figure 1 by investigating the loss profiles across noise levels of a diffusion model trained for 500 epochs on the CHEMBL228_Ki dataset. We find that there is a noise-dependent overfitting phenomenon, where losses may remain stable at high noise levels, but begin to increase at intermediate noise levels. While in practice this is mitigated by early stopping, this potential mismatch in learning progress across noise levels may lead to suboptimal training checkpoints. When training with fixed dropout rates, training loss is minimized with *no dropout* but leads to the most severe intermediate noise overfitting. When using an adaptive cosine dropout rate, training and validation losses are similar to the *no dropout* scenario at high noise, and comparable with a low-level of fixed guidance at intermediate noise, thus improving overall performance across the noise range.

2.3 Model Architecture and Joint Target-Feature Conditioning

The proposed architecture takes inspiration from the highly effective ADM [11] building blocks found in the original applications of diffusion to image generation. Given that our focus is on predictive models, we do not require a U-Net structure to decode latent vectors, and thus do not keep the associated noise-level skip connections. We additionally use Fourier embeddings to encode timestep information, following [31], and forego the channel-wise self-attention typically used at the end of each diffusion block. While typical conditional diffusion models incorporate the auxiliary information via the embeddings, our problem is slightly different in that we still require the network to learn good representations of the input features, as these provide much of the information required

for learning an accurate conditional score. Thus, the conditioning information from the features lie on the *main path* of the diffusion blocks, while the target to be denoised conditions the network via a joint embedding with the timestep. Our architecture allows for arbitrary feature extractor modules, supporting both molecular fingerprints and graph structures as feature representations. As the direct flow of information for the target values may still be useful for denoising, particularly at lower noise levels and for multitask problems, we introduce an optional gating mechanism in the residual path, controlling the flow of information from the target vector along this path. We report results of ablations on the architectural mechanisms in Appendix G and visual diagram for the architecture in appendix C.

3 Experiments

3.1 Calibrated Uncertainty

Diffusion models have two important characteristics indicating their potential applicability to uncertainty quantification. First, they make predictions by generating *samples* from $p(y|x)$, thus requiring no special accommodations to approximate the entire predictive distribution, rather than only its mean. Second, in contrast to models trained via maximum-likelihood, diffusion models make no assumptions about the conditional distribution. Thus, they have the capacity to model arbitrary, non-Gaussian likelihoods. We hypothesize that this should enable well-trained diffusion models to generate *calibrated* predictive distributions, a measure of the correspondence of the prediction sample quantiles with those of the true conditional distribution, and a key requirement for effective uncertainty quantification methods [21]. Intuitively, a well-calibrated model will contain the true target value within a 95% percentile interval of its samples approximately 95% of the time.

Table 1: Calibration MMD across datasets for different uncertainty methods.

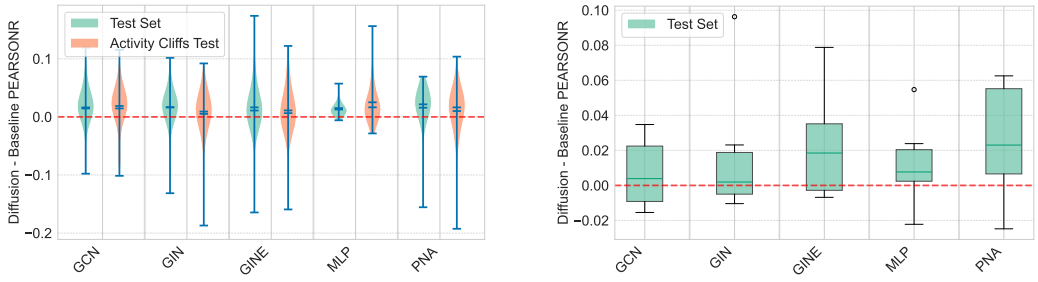
Method	ABL1	CASP3	CHEMBL228	CACO2	AqSolDB
Deep Ensemble[36]	0.275 ± 0.01	0.293 ± 0.01	0.38 ± 0.05	0.221 ± 0.04	0.263 ± 0.02
MC-Dropout[15]	0.405 ± 0.01	0.384 ± 0.01	0.53 ± 0.02	0.898 ± 0.13	0.804 ± 0.07
MVE[41]	0.425 ± 0.03	0.381 ± 0.02	0.501 ± 0.04	0.513 ± 0.08	0.399 ± 0.06
DiffPred	$\mathbf{0.062 \pm 0.0}$	$\mathbf{0.059 \pm 0.01}$	$\mathbf{0.168 \pm 0.05}$	$\mathbf{0.068 \pm 0.03}$	$\mathbf{0.047 \pm 0.01}$

In order to investigate this property, we propose the following experiment: For a given dataset, we generate 1000 prediction samples for each observation in the evaluation set. For each corresponding true target value, we note its percentile within the prediction samples, thus providing a sample from the empirical CDF with respect to the predictive distribution. If the model is well-calibrated, the empirical CDF should match the true target CDF and should be approximately $U \sim (0, 1)$. We map this to the quantiles of a standard normal distribution via the inverse quantile transform, and compare the obtained quantiles to the idealized quantiles of an exact $\mathcal{N}(0, 1)$ distribution. In order to obtain a quantitative measure of performance, we additionally computed the Maximum Mean Discrepancy (MMD) [19] to quantify the distributional divergence of the sample quantiles from the idealized Gaussian quantiles. After training to convergence, we observe that samples from DiffPred far exceed the calibration performance under baseline methods.

3.2 Representation-Agnostic Predictive Performance

In order to assess the potential benefits of leveraging score-based training for predictive performance, we compare our approach to modern MLP and graph neural network architectures where the feature extraction components of each model are fixed to the same number of layers and channels. For each dataset, we compare the relative performance difference observed across five random seeds. For baseline neural networks, we append an output MLP to the learned latent representations, where the dimensionality of the hidden layers is set to match the input dimensionality of each DiffPred block for the diffusion model (Figure 3).

Results. We find that training with diffusion robustly improved performance across most tasks for the MoleculeACE suite (see appendix I). In order to summarize the effect of training with diffusion across a range of architectures and datasets within each benchmark, we report aggregated results



(a) MoleculeACE aggregate

(b) TDC aggregate

Figure 2: Performance difference between DiffPred and baseline neural networks, aggregated across datasets, for various neural network architectures. The dotted red-line indicates a neutral performance threshold at zero. Training with diffusion improves performance *on average* across all architectures. Left: Aggregated performance across MoleculeACE datasets (excluding CHEMBL228_Ki), for both regular and activity cliff test sets. Right: Aggregated performance across TDC ADMET regression datasets (excluding CACO2_WANG).

Table 2: Multitask prediction results for DIFFPRED relative to GNN architectures trained via standard maximum likelihood. We report test set performance across five random seeds. The best performing method on each dataset is highlighted in green, and the second best in orange.

Model	**ADME-FANG** RMSE ↓	**Zinc12k** MAE ↓	**Tox21** AP ↑	**Tox21** AUROC ↑
MLP	0.558 ± 0.006	0.351 ± 0.006	0.436 ± 0.008	0.817 ± 0.017
GCN [32]	0.501 ± 0.012	0.084 ± 0.003	0.438 ± 0.007	0.837 ± 0.005
GIN [64]	**0.498 ± 0.01**	0.061 ± 0.009	0.43 ± 0.012	0.817 ± 0.003
GINE [25]	0.499 ± 0.004	**0.043 ± 0.002**	0.419 ± 0.014	0.829 ± 0.009
GPS [44]	0.502 ± 0.018	0.118 ± 0.033	**0.476 ± 0.01**	**0.843 ± 0.006**
PNA [10]	0.526 ± 0.019	0.08 ± 0.024	0.464 ± 0.018	0.833 ± 0.005
DIFFPRED	**0.489 ± 0.01**	**0.055 ± 0.002**	**0.487 ± 0.013**	**0.856 ± 0.004**

for each layer type in Figure 2. Predictive diffusion consistently provides performance benefits. We provide more granular results across individual datasets and architectures in appendix E.

3.3 Multitask Experiments

In the multitask regime, diffusion models are particularly interesting as they present a novel mechanism for joint training and inference. While neural networks trained with conventional maximum likelihood can learn to exploit dependencies between tasks, they are constrained to exploit these dependencies through a shared feature extractor $h_\theta(x)$ depending only on input features. Such constraints have led to various identifiability results in multitask learning [35, 9] demonstrating that learned representations in this setting can be reduced to specific, simple equivalence classes. Score-based models are thus more expressive, since predictions y_t for a given noise level are fully conditioned both by input features and by the noisy target values y_{t-1}, without further assuming any factorization on the label distribution. This property is particularly apparent where there is missing data, as predictions will be conditioned on imputed values from previous noise levels.

We verify the benefits of diffusion across three standard benchmarking datasets and report our results in Table 2. We performed basic hyperparameter tuning for all baseline models. For our diffusion approach, we perform no additional tuning, using the feature extraction parameters found for the corresponding baseline and keeping diffusion hyperparameters fixed. We report the best performing feature extraction component for each dataset as DiffPred. Across two of three benchmarks, DiffPred leads to improvements on test set metrics. This is particularly notable for Tox21, where the mode-coverage properties of diffusion models leads DiffPred to outperform in identifying the minority positive class.

4 Conclusion

Our exploration of diffusion models as a novel paradigm for training molecular property predictors indicates their significant potential in enhancing the performance and reliability of predictive models across real-world molecular datasets. We have motivated the introduction of a guidance schedule on the dropout rate *at training time* for the conditional feature information in order to address noise-dependent overfitting in data-limited regimes, while introducing novel architectural conditioning components. Our results indicate that in small-sample regimes, diffusion models may improve predictive accuracy and uncertainty calibration across a wide range of backbone architectures.

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A Limitations and future work

While we believe this work demonstrates that diffusion models offer many advantages over standard training in molecular property prediction, a clear limitation is additional computational cost. Since making predictions involves multiples samples, and each sample may require many forward passes, this significantly increases computational load at inference time. We note that for most settings in molecular machine learning, dataset sizes are not sufficiently large for this to be practically limiting. We also find that making predictions with only 10 to 50 sampling steps is effective. Predictive diffusion models are also likely to require longer training times, due to the amortized learning over many noise scales. Nonetheless, the novel mechanisms by which diffusion models generate predictions open up many research directions in supervised learning more broadly, particularly for multitask and transfer learning.

B Score-based Diffusion Models

We build on the continuous, score-based formulation of diffusion models in this work. Score-based diffusion models [53, 54] are a class of generative models which are trained to sample from a data distribution p_0 by learning an approximation to the noise-dependent score function [28] $\nabla_{x_t} \log p_t(x_t)$. The marginal densities p_t correspond to noisy snapshots of the data along a *forward-time* noising process of the form

$$dx_t = f(x_t, t)dt + g(t)dw, \tag{4}$$

where w corresponds to the Wiener process. Under suitable conditions for f and g, this stochastic differential equation admits a reverse-time SDE [1], which depends on the time-varying score

$$dx_t = (f(x_t, t) - g(t)^2 \nabla_{x_t} \log p_t(x_t))dt + g(t)d\overleftarrow{w}. \tag{5}$$

By taking the noising process to terminate at a sufficiently large timestep T, such that $p_T(x_T) \approx \mathcal{N}(x_T|0, \sigma_T^2\mathbf{I})$, it is thus possible to interpolate between data and Gaussian noise. Further, by training a model $s_\theta(x_t, t)$ to approximate $\nabla_{x_t} \log p_t(x_t)$ via a denoising score matching (DSM) objective [28, 58], we can obtain a generative diffusion model which can interpolate from noise to data by discretizing equation 5 and applying SDE solvers. Such models can be seen as generalizations of their discretized diffusion counterparts [24, 52], which are typically derived from a variational pespective.

C Model Architecture Diagram

We provide a detailed visualization of the architecture described in section 2.3.

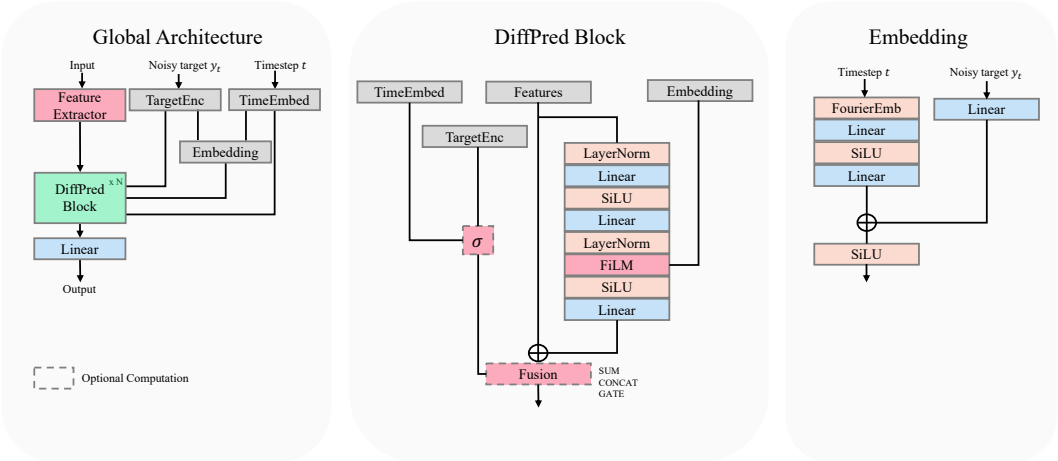


Figure 3: Overview of our proposed DiffPred architecture. Left: Global architecture. DiffPred can flexibly be integrated with arbitrary feature extractors. Center: DiffPred block, with an optional time-modulated gating mechanism allowing the target information to flow along the residual path. Right: Joint embedding vector for timestep and target vector.

D Related Work

A formulation of diffusion models for classification and regression tasks was initially introduced in CARD [22], where DDPMs are proposed to model general prediction tasks. However, their formulation relies on a diffusion process which terminates at the mean of a pre-trained predictor, thus requiring the diffusion model to effectively learn a denoising trajectory from a distinct endpoint for each data observation. In addition, the requirement of pre-training a discriminative model to initialize sampled predictions is practically cumbersome and obscures the contributions of diffusion to the final prediction. Our work demonstrates that with proper design choices, pre-training a discriminative model is not necessary for accurate predictions. More recently in computer vision, a trained ResNet classifier was combined with a text-conditioned diffusion model to correct uncertain predictions in image classification [38]. In [8], a robust generative image classifier is constructed from a pre-trained diffusion model via Bayes' theorem, under the assumption of a uniform prior over classes.

In scientific domains, diffusion models have increasingly been used to model differential equations, notably in weather prediction [43, 27] and fluid dynamics [39, 65], uncovering novel applications in forecasting and accelerated simulation. While these applications bear some relation to our work, in that the goal is to leverage diffusion to generate predictions about future states of a physical system, the formulation differs in key areas. Notably, diffusion models for PDE simulation maintain a primary focus on learning and simulating general dynamics, with the conditioning information playing a role in constraining these dynamics to specific observations, where this work focuses on maximizing the accuracy and contribution of the conditional score given high-dimensional features. Second, the observations will typically be less noisy, as they will correspond to direct measurements or observations from a physical simulator. As the stability of generated trajectories can degrade over time [40], sampling approaches that enable stable rollouts are often also the primary focus [50], rather than single-point predictive distributions.

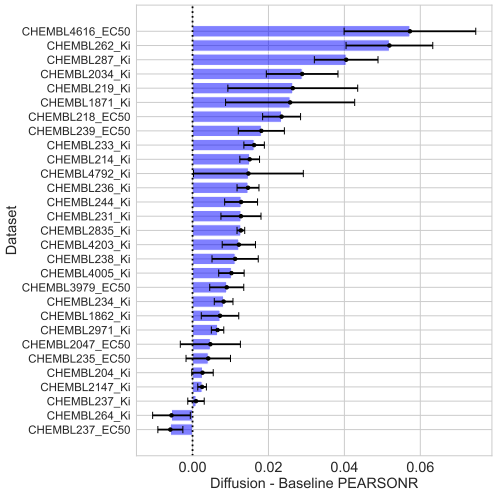
In chemistry and drug discovery, conditional diffusion models have been studied extensively in areas such as de novo protein design [20, 60, 62, 34], motif scaffolding [57, 66] and ligand design [49, 29]. However, applications of diffusion models to prediction in cheminformatics and drug discovery have to our knowledge remained relatively unexplored.

Within the context of molecular property prediction, prior attempts at improving predictive performance have focused primarily on graph representation learning, where there is a rich literature in expressive GNN architectures [32, 64, 25, 10, 48, 5, 4] and graph transformers [68, 67, 33, 44]. An alternative perspective on learning better representations was proposed in [55], where an infomax objective was used to enrich representations using both 2D and 3D molecular context. Our work positions itself as an orthogonal approach to these directions, enabling a novel training and prediction mechanism compatible with leading molecular representation learning architectures.

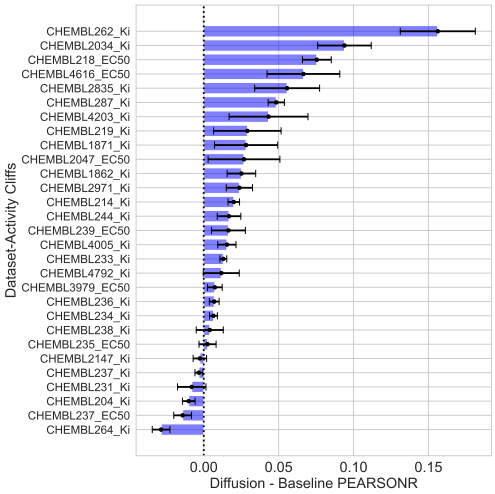
E Additional Results on Single-Task Experiments

We provide more granular single-task results than the aggregate presented in 3.2 by showing per-dataset improvements when training with diffusion. For CHEMBL, MLP models outperform GNN counterparts, whereas graph neural networks are superior for TDC datasets. CHEMBL results show performance differences for both regular and activity cliff test sets. In general, outperformance on the regular test set translates to outperformance on the cliff test set, and vice-versa. The magnitudes of the differences on cliff test sets are often larger.

E.1 MoleculeACE

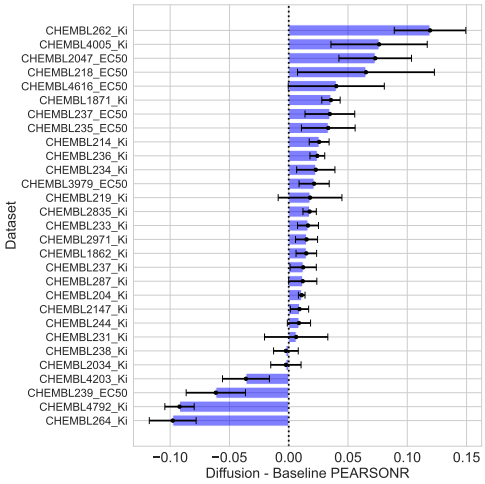


(a) MoleculeACE regular test sets - MLP

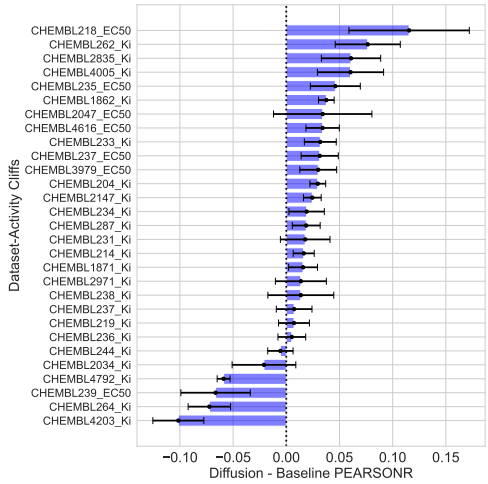


(b) MoleculeACE activity cliff test sets - MLP

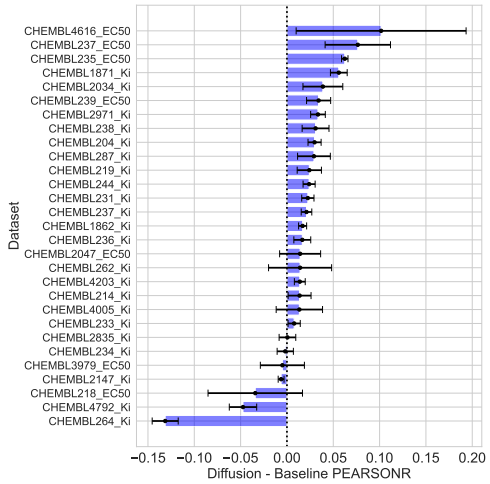
Figure 4: Comparison of the relative predictive performance of DiffPred with a standard neural network across CHEMBL datasets for the same feature extraction module. For fixed feature extraction modules, incorporating score-based training improves performance on the vast majority of datasets.



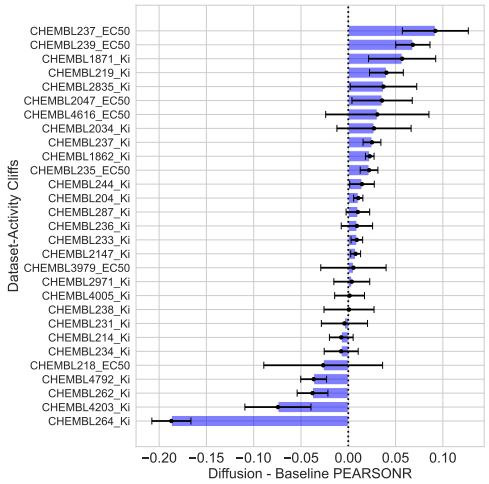
(a) Results on CHEMBL with GCN backbone - Regular Test Set



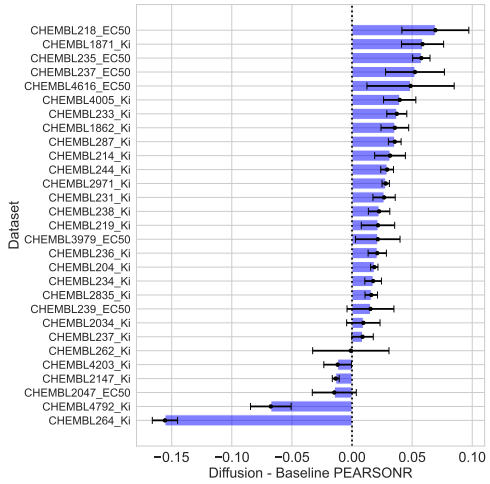
(b) Results on CHEMBL with GCN backbone - Cliff Test Set



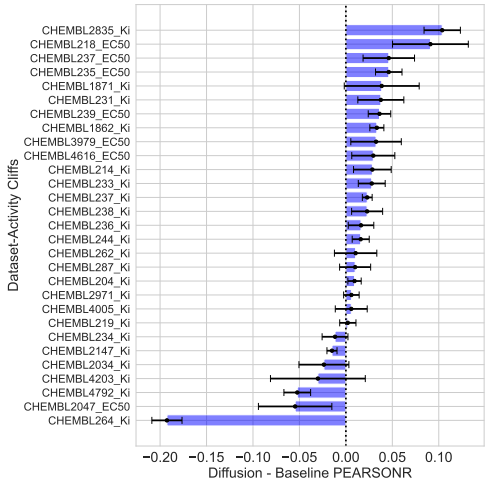
(a) Results on CHEMBL with GIN backbone - Regular Test Set



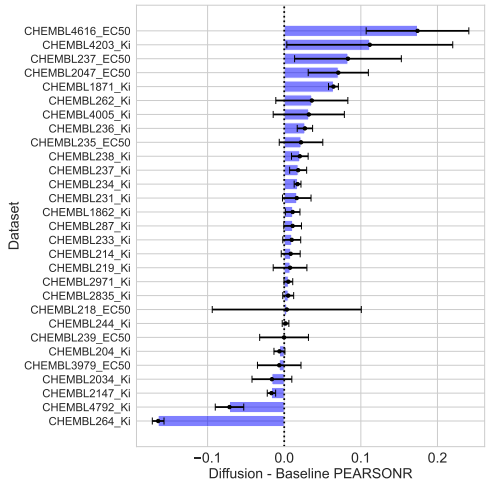
(b) Results on CHEMBL with GIN backbone - Cliff Test Set



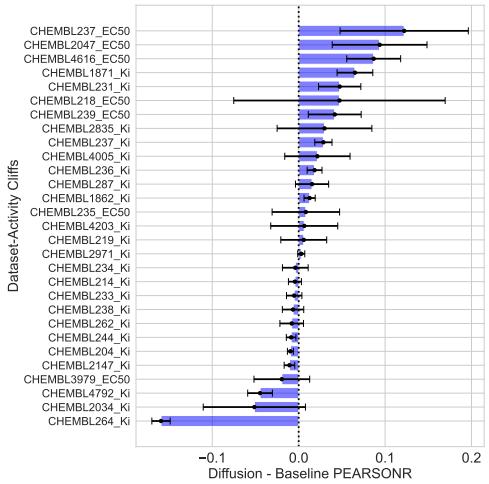
(a) Results on CHEMBL with PNA backbone - Regular Test Set



(b) Results on CHEMBL with PNA backbone - Cliff Test Set

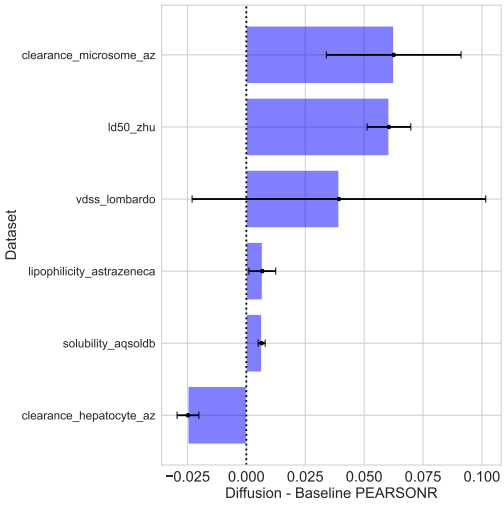


(a) Results on CHEMBL with GINE backbone - Regular Test Set

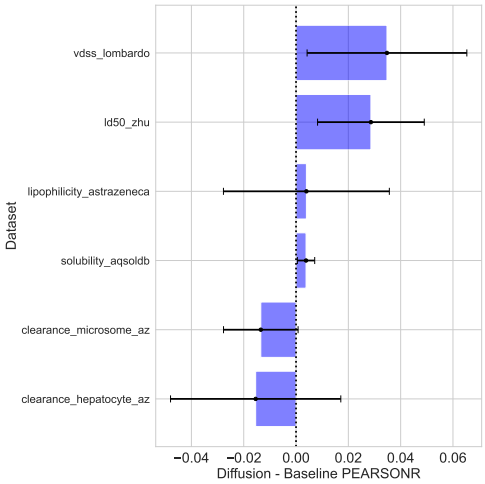


(b) Results on CHEMBL with GINE backbone - Cliff Test Set

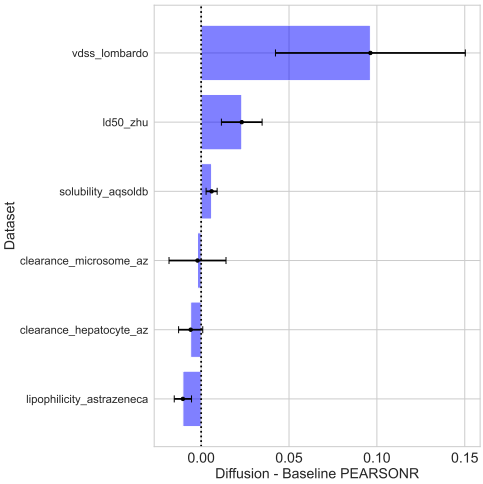
E.2 TDC ADMET



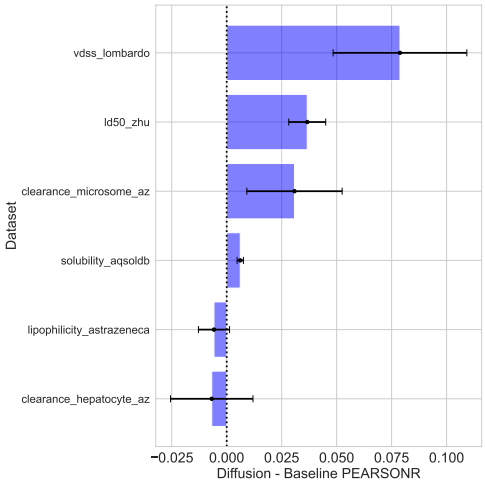
(c) Results on TDC datasets with PNA backbone



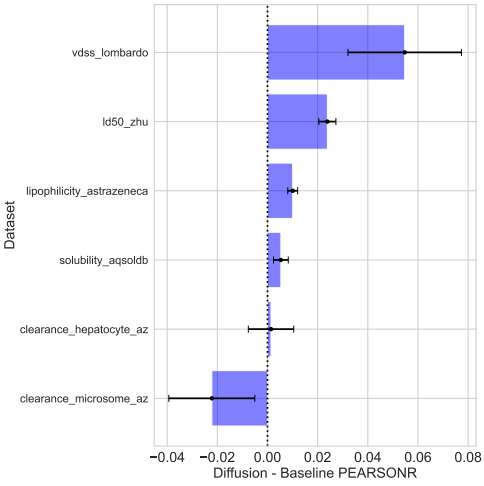
(a) Results on TDC datasets with GCN backbone



(b) Results on TDC datasets with GIN backbone



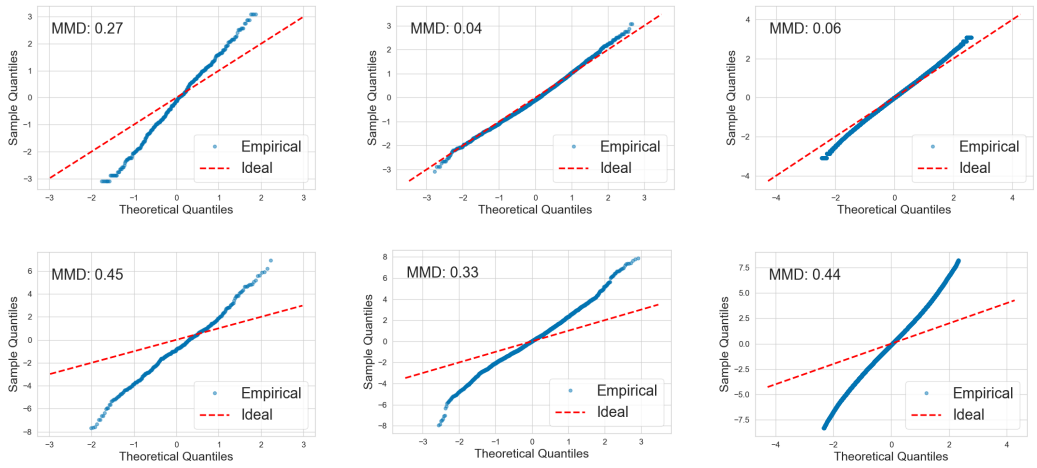
(a) Results on TDC datasets with GINE backbone



(b) Results on TDC datasets with MLP backbone

F Calibration Figures

We visualize individual calibration figures for Diffusion against MVE-based training for uncertainty calibration.



CHEMBL228 Aqsoldb ABL1

Figure 5: Test set calibration plots for the predictive distribution of a DiffPred model and MVE neural network on CHEMBL228_Ki, Aqsoldb and Dockstring [17] task ABL1 for a particular random seed. An MMD metric is computed with respect to the theoretical Gaussian quantiles. Top: DiffPred calibrations. Bottom: MVE calibrations.

G Architectural Conditioning Details

G.1 Details on Target Fusion and Gating Mechanism

We detail the exact architectural mechanism for the fusion of target information along the main path of our diffusion network. The Fusion operation in 3 refers to either a concatenation, summation or GRU-like activation update. The candidate vector $\mathbf{h}_t$ for the target information update is obtained as

$$\mathbf{h}_t = \sigma(T_\phi(\mathbf{e_t}))\tilde{\mathbf{y}_t} \tag{6}$$

where σ is the sigmoid activation function, T_ϕ is a linear transformation of the target embedding $\mathbf{e_t}$, and $\tilde{\mathbf{y}_t}$ is the target encoding at timestep t. The product is a scalar product rather than a channel-wise scaling, which simplifies the computation to a scaling of the entire target and feature representations. For sum and concatenation operators, this candidate representation is directly fused with the feature vector $\mathbf{g}_t$. Denoting the weight given to the target vector as $z_t = \sigma(T_\phi(\mathbf{e_t}))$, the gating mechanism is applied to the candidate vector $\mathbf{h}_t$ and the feature branch representation $\mathbf{g}_t$, where the final output of the fusion operation is given by the linear interpolation

$$\mathbf{g}'_t = z_t\tilde{\mathbf{y}_t} + (1 - z_t)\mathbf{g}_t \tag{7}$$

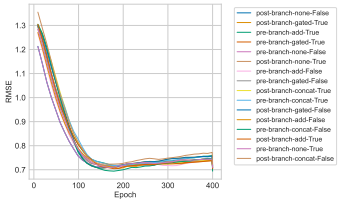
This mechanism enables the model to learn a timestep-dependent fusion of the target information with the feature representation, thus allowing the model to prioritize the target information dynamically throughout the diffusion process. Intuitvely, we would expect this information to be most useful at lower noise levels, where the model is more certain about the target information, and the next timestep prediction sample should only slightly deviate from the current state.

G.2 Ablation Experiments on Fusion of Target and Feature in the architecture main path

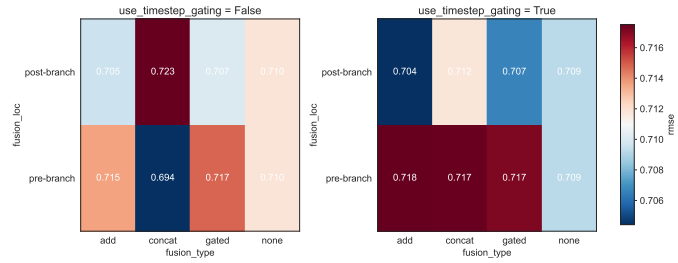
We compare the validation performance throughout training and after early stopping on different architectural configurations. For each fusion type of the target information (no fusion, summation, concatenation and gating), we compare results when doing the fusion *before* the residual branch containing embedding information is merged vs. after it is merged (pre-branch vs post-branch). We also compare this with and without the modulation of the target vector via a learned timestep-dependent scalar scaling factor between 0 and 1. Note that the matrix contains certain redundancies,

for example for no fusion, there is no difference between pre/post-branch or with the use of timestep gating. For the gated fusion mechanism, the timestep modulation is always used. Any minor differences is attributed to random stochasticity across training runs.

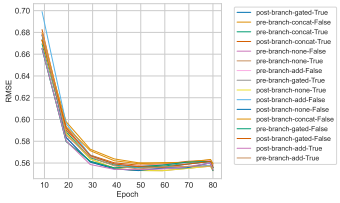
G.2.1 Single-Task Datasets



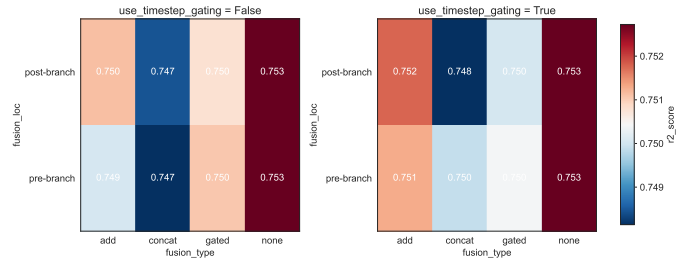
(a) VAL RMSE during training on CHEMBL with various fusion strategies.



(b) Final VAL RMSE on CHEMBL with various fusion strategies.

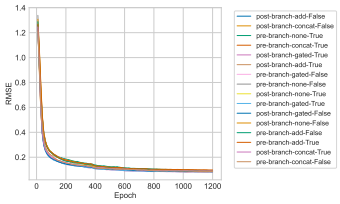


(a) VAL RMSE during training on dockstring ABL1 with various fusion strategies.

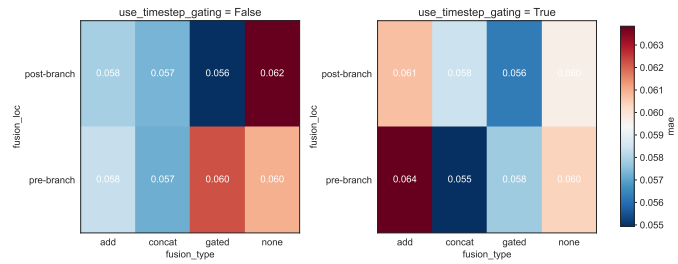


(b) Final VAL RMSE on dockstring ABL1 with various fusion strategies.

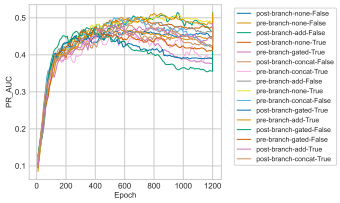
G.2.2 Multitask Datasets



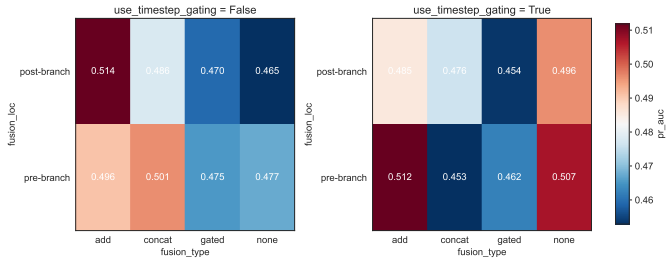
(a) VAL RMSE during training on ZINC12k with various fusion strategies.



(b) Final VAL MAE on ZINC12k with various fusion strategies.



(a) VAL PRAUC during training on Tox21 with various fusion strategies.



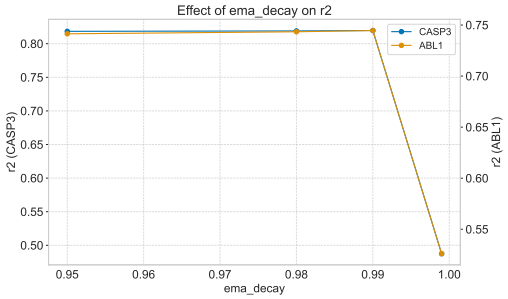
(b) Final VAL PRAUC on Tox21 with various fusion strategies.

H Ablations on Diffusion Hyperparameters

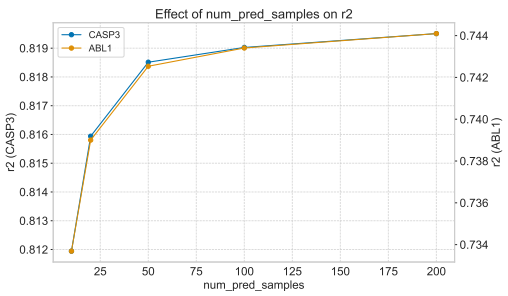
We conducted a series of ablations on the test performance sensitivity to changes in common diffusion hyperparameters. We performed these ablations for larger datasets where the targets are generated systematically through the automated docking suite provided in [17]. We used the ABL1 and CASP3 targets, each containing about 250K observations. We further performed this analysis on smaller scale datasets across ADMET (CACO2) and activity (CHEMBL228_Ki). Because we used these ablations to set default hyperparameters, these two datasets are not included in the main results of section 3.2. In general, results on the Dockstring datasets align with observations in large unconditional image generation tasks, however results are noisy on the small-scale datasets. In general, there are clear trends for EMA decay, number of prediction samples, guidance schedule used and minimum noise level σ. For other parameters however, performance is data and architecture dependent, and test metrics are generally less sensitive to the exact settings. The test set R^2 score for each dataset is plotted on a separate y-axis for each figure.

H.1 Ablations on large-scale docking datasets

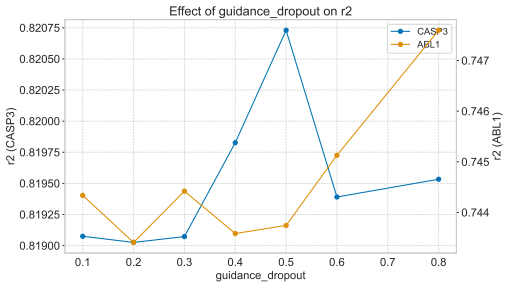
Ablations on docking datasets ABL1 and CASP3 use an MLP feature extractor module.



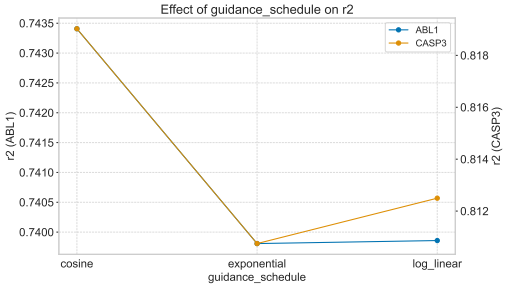
(a) EMA Decay



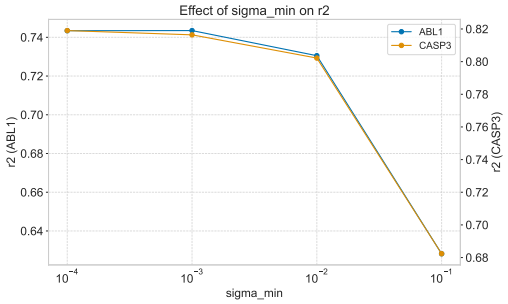
(b) Number of Prediction Samples



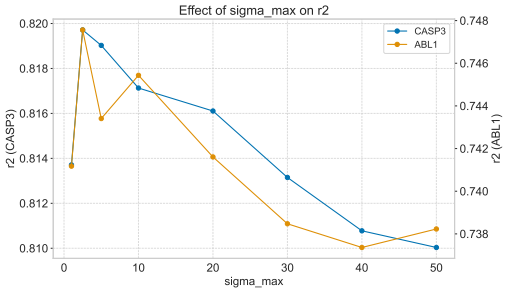
(a) Guidance Dropout Max



(b) Guidance Schedule



(a) Sigma min



(b) Sigma max

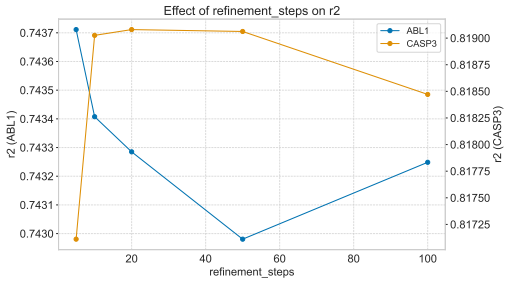
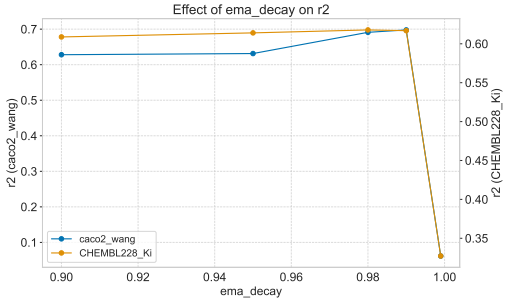


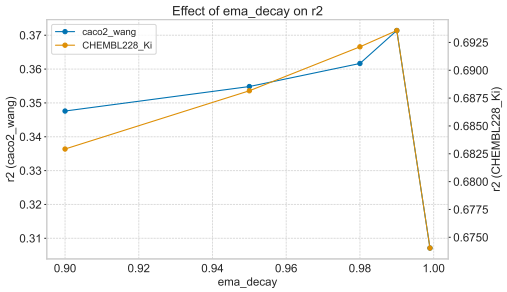
Figure 6: Number of Sampling Steps

H.2 Ablations on small real-world datasets

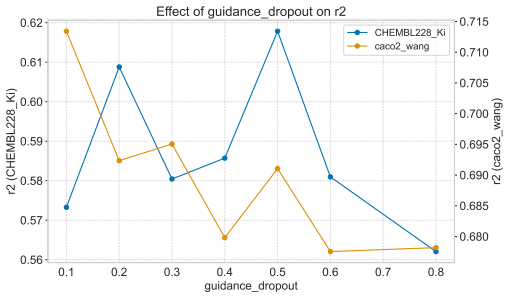
For the small scale datasets, we provide results for both an MLP and GIN feature extractor, given that each approach is markedly better for one type of dataset.



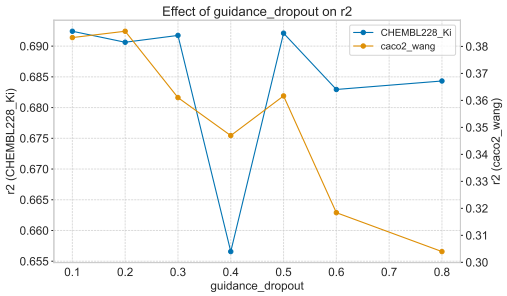
(a) EMA Decay GIN



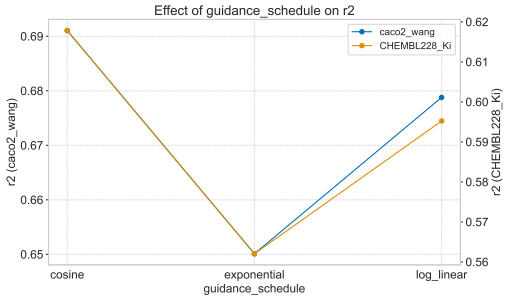
(b) EMA Decay MLP



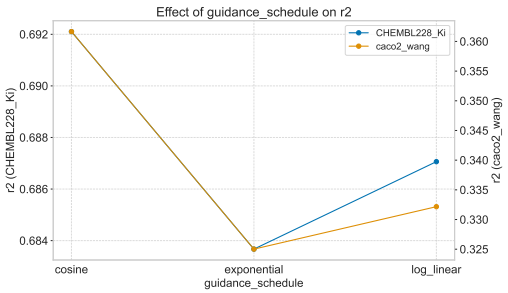
(a) Guidance Dropout Max GIN



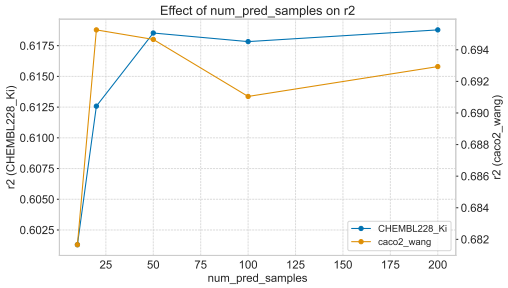
(b) Guidance Dropout Max MLP



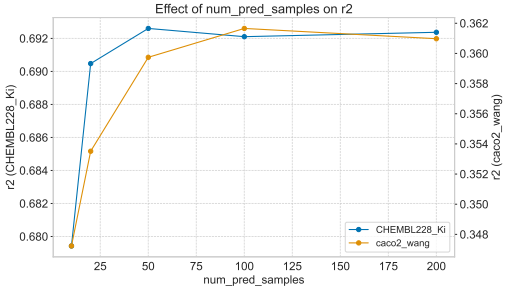
(a) Guidance Schedule GIN



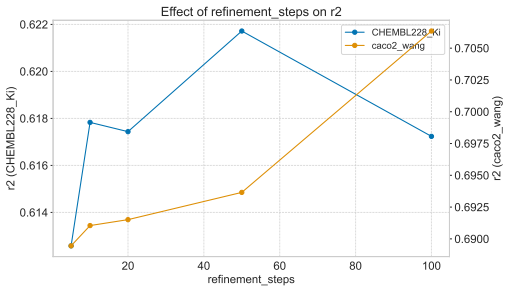
(b) Guidance Schedule MLP



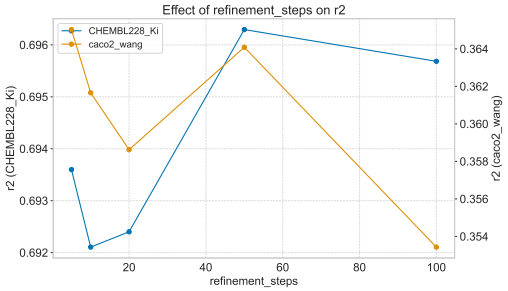
(a) Number of Prediction Samples GIN



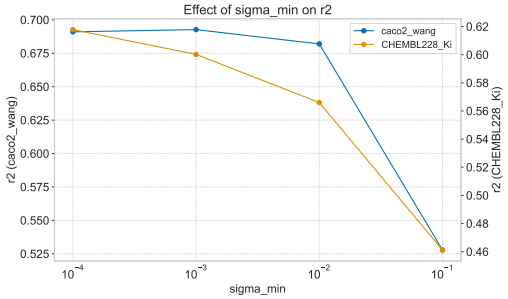
(b) Number of Prediction Samples MLP



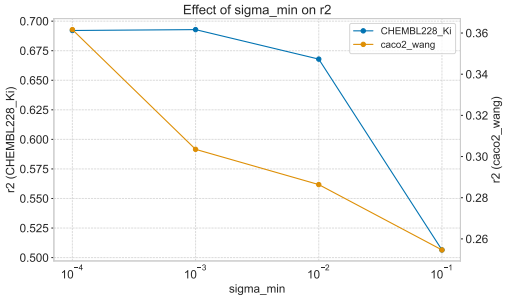
(a) Number of Sampling Steps GIN



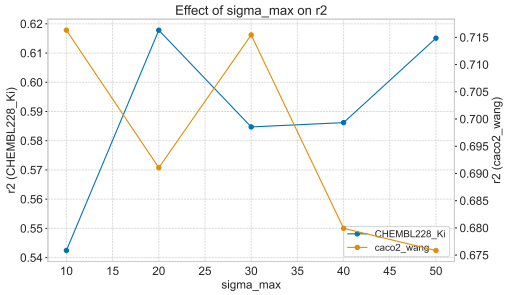
(b) Number of Sampling Steps MLP



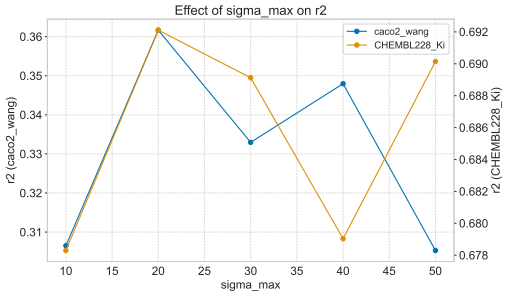
(a) Sigma min GIN



(b) Sigma min MLP



(a) Sigma Max GIN



(b) Sigma Max MLP

H.3 Conditional vs. Unconditional Training Loss Profiles

We highlight the difference in typical loss profiles observed for predictive diffusion models relative to standard loss profiles in the unconditional case. This motivates a more uniform prioritization of the loss across noise levels, in contrast to one focused on intermediate noise regimes.

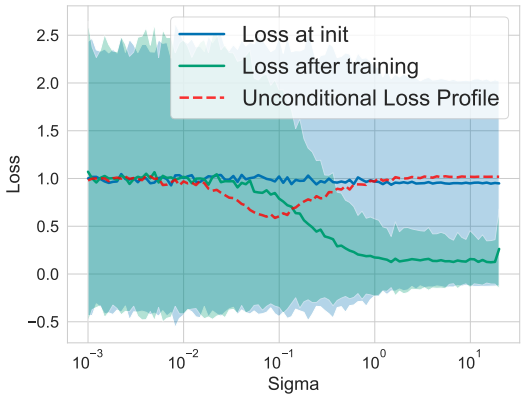


Figure 7: Loss profiles across noise levels for predictive diffusion (green curve) on CHEMBL228_Ki, along with a typical loss profile for an unconditional diffusion model trained on the classic two moons dataset.

I Experimental Details

I.1 Implementation

For all our experiments, fully-connected networks are trained using a concatenated RDKIT [37] and ECFP4 [46] fingerprint feature representation with a total length of 4000 bits. For graph neural network models, we featurize directly the SMILES representation for each molecule via the default atom and bond featurizers in [42]. Experiments were performed using Graph Convolutional Networks (GCN) [32], Graph Isomorphism Networks (GIN) [64], GINE [25], PNA [10], and GPS [44] for larger scale multitask datasets, with their implementations provided by Pytorch Geometric [14].

I.2 Data

For all multitask benchmarks, we use the test splits provided by the Polaris repository [61]. For single-task benchmarks, we use the test splits associated with the associated benchmark suite for both MoleculeACE and TDC datasets. Validation splits use a random split strategy, where a random seed is used for all experiments.

I.2.1 Single-Task Datasets

We performed our single-task experiments on both activity prediction tasks and ADMET prediction tasks in order to assess results across a wide range of applications. For activity prediction, we reported results on the MoleculeACE [56] suite of benchmark datasets, which represents a curated selection of datasets from the ChEMBL bioactivity database [18]. Additionally, this benchmark suite provides an alternative *activity cliff* test set, where the original test set is restricted to molecules for which there exist highly similar structures in the training set with large differences in bioactivity values. For ADMET tasks, we compare performance on six benchmark regression datasets from the Therapeutics Data Commons repository [26], covering a diverse range of pharmacokinetics and toxicity prediction.

I.2.2 Multi-Task Datasets

Zinc12k was originally a single-task dataset introduced in [12] to provide a sufficiently large, relatively noiseless, benchmark for evaluating graph neural networks reliably. Because the targets for this dataset correspond to aggregating computed $logP$ and synthetic accessibility scores, [2] introduce a multitask version of this dataset where the objective is to regress the constituent scoring functions simultaneously. Tox21 [45, 63] is a widely used multilabel binary classification dataset consisting of 12 toxicity endpoints, with approximately 20% missing labels and heavily biased to the positive class. Due to our interest in identifying toxic compounds in the presence of class imbalance, we report average precision (AP) in addition to AUROC values typically used for evaluation. In order to generate predictions with diffusion in the classification setting, we binarize labels to values in $\{-1, 1\}$, standardize them and train analogously to the regression setting. Final predictions at inference time are generated via a thresholded logistic function. ADME-FANG [13] is a recently introduced, smaller but highly realistic multitask benchmark of real-world data for pharmacokinetics.

I.3 Hyperparameters

I.3.1 Diffusion Hyperparameters

Default hyperparameters for DiffPred were set according to the ablation studies in section H, where reasonable values were selected for hyperparameters where the performance was not sensitive to the exact value. Compromise was made for defaults between small and large datasets, as such it is reasonable to expect that the optimal hyperparameters for a given dataset may be different and that performance could be further improved. The default hyperparameters are shown in table 3. Hyperparameters specific to the architectural conditioning mechanism are shown in table 4, and were set to reasonable defaults according to the ablation studies in section G.1.

Hyperparameter	Value
Number of sampling steps	10
Number of prediction samples	200
EMA decay	0.99
Guidance dropout max	0.2
Guidance schedule	cosine
Sigma min	1e-4
Sigma max	20

Table 3: Default hyperparameters for DiffPred across all experiments.

Hyperparameter	Zinc12k	ADME-Fang	QM9	Tox21	Single-Task
Target fusion type	Gating	Gating	CONCAT	SUM	None
Target fusion pre/post branch	Post	Post	Post	Pre	N/A
Target modulation	Yes	Yes	Yes	Yes	N/A

Table 4: Hyperparameters specific to the architectural conditioning mechanism for DiffPred. Fusion Type options are sum, concat, gating and none. Pre/post branch refers to whether the target information is fused before or after the residual branch containing the embedding information. Target modulation refers to whether a learned timestep-dependent scalar is used to modulate the target vector at each timestep.

I.3.2 Architectural and Training Hyperparameters

For all single-task experiments in 3.2, hyperparameters were fixed to default values where the feature extractor architecture was fixed for both DiffPred and baseline neural networks. Output layers were set to similar dimensionality such that the capacity of the models was similar. Default hyperparameters are summarized in table 5. For all experiments, we use an Adam optimizer with a learning rate of 1e-3, a learning rate scheduler set to reduce the learning rate by a factor of 0.5 after a patience of 10 epochs without validation improvement, and a batch size of 1024.

Hyperparameter	Value
Number of epochs	800
Batch size	1024
Batch Norm	False
Dropout	0
Feature extractor hidden dims	[256, 128]
Hidden dimension	512
Output MLP hidden dim	128
Output MLP layers	3
Activation	SiLu

Table 5: Default hyperparameters for all architectures across single-task experiments. Hidden dimensions for PNA were the exception and were set to [70, 70, 70, 70] following default settings from the original paper.

For multitask experiments, we used a larger batch size for QM9 of 5000. For Zinc12k, Tox21, and ADME-Fang, a random search with 50 trials and median pruner (implemented via Optuna) was used to find the best hyperparameters. For QM9, a smaller sweep of 10 trials was run due to computational constraints and dataset size. The search space was set to sweep over feature extractor hidden dimensions, number of output MLP layers, output MLP hidden dimensions, dropout rate, batch norm and the presence of a residual connection in the output MLP module.

I.4 Computing Resources

All experiments were conducted on a single A5000 GPU. Each training run was dispatched on a single worker via the Pytorch-lighting accelerator abstraction. Experiments were dispatched in parallel in most cases, as such precise runtimes for each experimental result were not calculated. However, a single training run for the smaller single-task datasets, over 600 epochs, can be completed in under 10 minutes for all feature extractor architectures. For the largest datasets, such as QM9, a single full training execution over 300 epochs ran in approximately 90-120 minutes.